

Midterm

Phil 305 — Propositional and Modal Logic

Part 1: S5

Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$
2. $\Diamond\Box p \rightarrow \Box\Diamond p$
3. $\Diamond(\neg p \vee r) \rightarrow (\Box(p \vee q) \rightarrow \Diamond(q \vee r))$
4. $(\neg p \vee \neg\Box\Box p) \rightarrow \neg\Box p$

Part 2: K

Use the simplified rules for **K** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **K**.

5. $\Diamond$

Part 3: S4

Use the simplified rules for **S4** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S4**. Remember **S4** is the logic **KT4**, so it has the rules for **T** and **4** as well as **K**

8. $\Box(\Box p \rightarrow q) \vee \Box(\Box q \rightarrow p)$